

Vibin Ramakrishnan Ph. D. FRSC FICS FRSB

Professor, Department of Biosciences & Bioengineering &
Mehta School of Data Science and Artificial Intelligence

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EDUCATION AND EMPLOYMENT

- Professor, Indian Institute of Technology, Guwahati – 781039. India (2019 -)
- Associate Professor, Indian Institute of Technology, Guwahati – 781039. India (2015-2019)
- Assistant Professor, Indian Institute of Technology, Guwahati – 781039. India (2011-2015)
- Faculty Scientist; Institute of Bioinformatics and Applied Biotechnology (IBAB), Bangalore.
- Post-Doctoral Research Associate: RPI, Troy NY - 12180 USA
- Ph.D.: Indian Institute of Technology, Bombay, Powai-400076 India.

RESEARCH INTERESTS: Network Medicine, Drug Delivery vehicles, Computational Biology

AWARDS & SCHOLASTIC ACHIEVEMENTS

- **FRSC** Fellow of Royal Society of Chemistry UK
- **FICS** Fellow of Indian Chemical Society
- **FRSB** Fellow of Royal Society of Biology
- **IYBA Fellow:** Innovative Young Biotechnologist Award Extension (IYBA 2011-12), Dept. of Biotechnology, Ministry of Science and Technology, Govt. of India.
- **IYBA Fellow:** Innovative Young Biotechnologist Award (IYBA 2007-08), Dept. of Biotechnology, Ministry of Science and Technology, Govt. of India.
- Post-Doctoral Fellowship funded by National Science Foundation United States
- Cochin University Merit Scholarship by CUSAT for top two students in each discipline

PROFESSIONAL AFFILIATIONS

National Academy of Sciences. India
Biophysical Society
Controlled Release Society of India
Protein Society

American Chemical Society
American Peptide Society
Society of Biological Chemists
The International Society for Computational Biology (ISCB)

EDITORIAL BOARDS

1. [Journal of Controlled Release](#), Publisher: Elsevier, Impact Factor: 11.5
2. [Scientific Reports](#), Publisher: Nature Publishing Group, Impact Factor: 4.6 (2018-2024)
3. Amino acids, Publisher: Springer, Impact Factor: 3.0

PUBLICATIONS

BOOKS

1. De novo Peptide Design; Principles & Applications
Publisher: Elsevier ISBN: 978 -0-323 - 99917-5. Publisher: Elsevier
2. Biophysical Characterisations of Peptides
Publisher: Springer ISBN: 978-1-0716-3404-2. Publisher; Springer

Research Papers + Book Chapters	Books	Patents (International, Indian)
59 + 22 = 81	2	1, 13

Sponsored Research Projects (Completed)	Sponsored Research Projects (Ongoing)	Computational Tools Developed
12	2	5

PhD Thesis	M.Tech Thesis	B.Tech Dissertations
Defended (12) Ongoing (6)	Completed: 14 Ongoing: 1	14

IMPORTANT PUBLICATIONS

1. Ruchika Goyal, Gaurav Jerath, Akhil R., Aneesh Chandrasekharan, Eswara Rao Puppala, Srikanth Ponneganti, Anupam Sarma, V.G.M. Naidu, T. R. Santhoshkumar, and Vibin Ramakrishnan*. "Geometry Encoded Functional Programming of Tumor Homing Peptides for Targeted Drug Delivery", **Journal of Controlled Release** (2021) DOI: 10.1016/j.jconrel.2021.03.0101 **Impact Factor: 11.467. Publisher: Elsevier**
2. Gaurav Jerath, Pramod Darvin, Yvonne Christian, Vishal Trivedi, T. R. Santhosh Kumar and Vibin Ramakrishnan*. Delivery of Small Molecules by Syndiotactic Peptides for Breast Cancer Therapy. **Molecular Pharmaceutics** (2022) DOI: acs.molpharmaceut.2c00238 **Impact Factor: 5.364. Publisher: American Chemical Society**
3. Sasidharan, Sayandeep Ghosh, Rishi Sreedhar, Kalpana Kumari, Subhash Thota, and Vibin Ramakrishnan* Anisotropic Ferromagnetic Organic Nanoflowers, **Journal of Physical Chemistry C** (2022) DOI: 10.1021/acs.jpcc.2c01462. **Impact Factor: 4.177. Publisher: American Chemical Society**
4. Ruchika Goyal, Gaurav Jerath, Aneesh Chandrasekharan, Yvonne Christian, T. R. Santhoshkumar, and Vibin Ramakrishnan*, Molecular hybridization combining tumor homing and penetrating peptide domains for cellular targeting, **Drug Delivery and Translational Research** (2021) DOI : 10.1007/s13346-021-01035-z. **Impact Factor: 5.671. Publisher: Springer**
5. Jahnu Saikia, Gaurav Pandey, Sajitha Sasidharan, Ferrin Antony, Harshal B. Nemade, Sachin Kumar, Nitin Chaudhary, and Vibin Ramakrishnan*. Electric Field Disrupts Amyloid Aggregation; Potential Non-invasive Therapy for Alzheimer's Disease. **ACS Chemical Neuroscience**. (2019), doi:10.1021/acchemneuro.8b00490. **Impact Factor: 5.78. Publisher: American Chemical Society**
6. Gaurav Jerath, Ruchika Goyal, Vishal Trivedi T.R. Santhoshkumar and Vibin Ramakrishnan*. Syndiotactic Peptides for Targeted Delivery. **Acta Biomaterialia**. (2019), doi: 10.1016/j.actbio.2019.01.036. **Impact Factor: 10.663. Publisher: Elsevier**
7. Rahul Metri, Gaurav Jerath, Govind Kailas, Adityabarna Pal & Vibin Ramakrishnan*. Structure Based Barcoding of Proteins. **Protein Science** (2014) 23, 117-120. **Impact Factor: 6.725. Publisher: Wiley**
8. Vibin Ramakrishnan; Ranjit Ranbhor and Susheel Durani. Existence of Specific "Folds" in Polyproline-II Ensembles of an Unfolded Alanine Peptide Detected by Molecular Dynamics. **Journal of the American Chemical Society** (2004) 126, 16332-16333. **Impact Factor: 15.420. Publisher: American Chemical Society**